SUPERMARKET BILLING SYSTEM

21CSC101T – Object Oriented Design and Programming

Mini Project Report

Submitted by

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BONAFIDE CERTIFICATE

Certified that this Course Project Report titled **"Supermarket billing system in C++** is the bonafide work done by **GAURI GUPTA[RA2211026010359], ADITYA SHARMA[RA2211026010295], KHUSHI UPADHYAY[RA2211026010312], NEELANSH BHARGAVA[RA2211026010360]** who carried out under my supervision. Certified further, that to the best of my knowledge, the work reported herein does not form part of any other work.

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ABSTRACT

SUPERMARKET BILLING SYSTEM

The invoicing system **of** the **store includes** automation **in the** supermarket. This software will help **vendors manage** various types of **information abou t customers. This** product will help **users** work in a **good** environment. It **re quires a lot of** time and energy that could be **used for better work. In additi on, as** the **power** of **customers increases,** managing **each customer's** info rmation **becomes** a **tedious** task.

In the **guide, sellers face many inefficiencies. Data storage** is one of the **most important issues. Writing the perfect user experience report** is **diffi cult.** Large **files should** be **stored** where **important** and irrelevant **data sho uld** be **stored. Automate** all **these** problems and **try** to **eliminate** them in th e **most appropriate way.**

The new system will **meet** the **needs** of **every seller in** the **market** so that **th e seller** can manage the system **effectively. The purpose of the developm ent of the supermarket** is **to make** the system **more** reliable, easier, **faster** and **richer in information.**

CHAPTER 1

INTRODUCTION

Supermarket billing system is the system to automate the process of ordering and billing of a supermarket store. Supermarket is the place where customers come to purchase their daily using products and pay for that. So there is a need to calculate how many products are sold and to generate the bill for the customer.This system is built for fast data processing and bill generation for supermarket customers. It also allows the customer to purchase and pay for the items purchased. The users will consume less time in calculation and the sales activity will be completed within a fraction of seconds whereas in a manual system will make the user to write it down which is a long procedure and it also consumes a lot of time. Because of this software, paper work will be reduced and the user can spend more time on monitoring the supermarket. The project will be user friendly and easy to use. This project is helpful to computerize the bill report and generating the items details. The billing data is a vast collectionof product name, price and other product specific data. A product when billed is searched and its price is added to the bill based upon the product quantity. The system also contains discounts on various products so that the product is offered at discounted price while billing. The supermarket billing system is built to help supermarkets calculate and display bills and serve the customer in a faster and efficient manner. This software project consists of an effective and easy GUI to help the employees in easy bill calculation and providing an efficient customer service. With the continuous development and improvement of computer technology, communication technology, network technology, scale database technology, the commercial supermarket has become a developing technology worldwide.

FEASIBILITY STUDY

Feasibility study is a process of analyzing and identifying if a problem can be

solved or not solved, focusing on helping answer the essential question of “should we continue the proposed project ideas?” All activities of the study are directed toward helping answer this question. A Feasibility Study is generic in nature and can be applied to any type of project, be it for systems and software development, making an acquisition, or any other project. Feasibility Study is a test of the system according to its workability, impact of the organization, ability to meet user needs and effective use of the resources. We can test our system by different type of the feasibilities. There are basically six parts to any effective Feasibility Study:

(1) The Project Scope -which is used to define the business problem and/or opportunity to be addressed. The old adage, "The problem well stated is half solved," is very apropos. The scope should be definitive and to the point; rambling narrative serves no purpose and can actually confuse project participants.

(2) The Current Analysis -is used to define and understand the current method of implementation, such as a system, a product, etc. From this analysis, it is not uncommon to discover there is actually nothing wrong with the current system or product other than some misunderstandings regarding it or perhaps it needs some simple modifications as opposed to a major overhaul.

(3) Requirements - how requirements are defined depends on the object of the project's attention. For example, how requirements are specified for a product is substantially different than requirements for an edifice, a bridge, or an information system.

(3) The Approach- represents the recommended solution or course of action to satisfy the requirements.

(4) Evaluation - examines the cost effectiveness of the approach selected. This begins with an analysis of the estimated total cost of the project. (6) Review - all of the preceding elements are then assembled into a Feasibility Study and a formal review is conducted with all parties involved.

There are 5 types of the feasibilities which are discussed here. These are as follows:

1. **Technical Feasibility**: A study of resource availability that may affect the ability to achieve an acceptable system. This evaluation determines whether the technology needed for the proposed system is available or not. This system can be made in any language that support good user interface and easy database handling.

Technical needs may include:

Front-End Selection: Front-End means a language that is used for user interface designing and coding. Front-End should have following qualities:

- It must have a graphical user interface that assist employees that are not from some IT background.
- Scalability and Extensibility
- Robustness
- According to the organization requirements and culture.
- Must provide excellent reporting features with good printing support Platform independent.
- Easy to deploy and maintain.
- Event driven programming.

Economic Feasibility:

In this we consider following tasks:

- The cost to conduct a full system investigation.
- The cost of hardware and software for class of application being considered.
- The benefit in the form of the reduced cost.
- Our system has a lot of features at a minimum cost so it is feasible to implement and it will be very much beneficial to the sellers in the reduced

cost. It's software and hardware cost is also low then the existingsystem.

2. Operational Feasibility:

In this feasibility we consider following points:

In the new system we made some major changes for the staff members so that they have to be trained to use the newly added facilities. These major changes are possible and give a new era in the Supermarket in production and sales management.

3. Schedule Feasibility:

Time evaluation is most important consideration in development of the project. So the project is concerned should be completed with fixed in scheduled time as far as company is concerned. New system is not so much big so it is easy to make in few days.

1.2 SCOPE OF THE WORK

Scope of this project is to investigate and design a software solution which can facilitate both customer and salesperson in performing their daily tasks, improving efficiency, and helping them to be more productive. This project will provide a solution through which salesperson can easily manage, handle and generate all required information in their respective format when needed. It provides quick way of operation by capturing the manual process and automating them. It will help them to manage the bill details, financial data, andhistorical data and also in producing documents of different formats for different customers. This solution will help salesperson in reducing effort spend on managing many bills. It will also provide them opportunity to explore possibility of generating documents, managing financial details. This system will help the salesperson to manage in fast billing. It will help to maintain the data of the purchased items. It also gives bill to the customers. It will set the rates of taxes and commission on products. The project will enable to see the report regarding

product and category in a fixed period of time. It can also change the Graphical User interface of the system.

1.3 PROBLEM STATEMENT

In today's fast paced society, it's very hard to be competitive without using cutting-edge technology available in market. After years of business, the data has grown much. It is becoming a challenge for person to manage that data in an effective way. To be more productive in order processing, he needs a solution which can facilitate their current processes with use of technology and software. With increased amount of orders, it is becoming difficult for salesperson to manage orders in effective and efficient manner. It is very hard to go through all paper work and backtracking orders. If there is any complain or review of any order, it takes large amount of effort and time to backtrack and fix the problem. This results in loss of resources, increased time, and low output.

Drawbacks of Manual System (Current System):

- Time consuming: Getting the required information from the available data takes a lot of time. Changing, editing and updating the information contained in several files are a slow and time-consuming process.
- Poor communication: A manual system requires employees and managers to write down each time an item is removed from the inventory. If one employee forgets to mention that the last coffee producthas been removed from the inventory the admin or manager expects the item to still available for a customers during sale.
- Need of Effort: In manual system, an Item's record is maintained in separate files so it takes much effort to collect data from several Stores for and if we want to change or delete the data of any transaction then it has to be changed or deleted from all the files and places it stored.
- Needs Large Space: In manual work done data item has to be stored at several places, similarly student's record is maintained in separate registers. It requires more storage space.

1.4 AIM AND OBJECTIVES OF THE PROJECTAIM:

To make software fast in processing, with good user interface so that user can change it and it should be used for a long time without error and maintenance.

OBJECTIVES:

- To reduce the time for organisation.
- To increase efficiency and accuracy of the system.
- To reduce pressure on the labour and relieving man power from repetitive and dull job.
- To make the retrieval of information faster.
- To make the system more feasible
- To reduce large amount of paper work.
- To make the system more reliable and to avoid any ambiguity.
- To reduce the cost factor of the system.
- To make the system more flexible.
- It provides a convenient solution of billing pattern.
- It maintains new entered category or products.

Description of the Existing System :

Many Supermarkets use this type of billing system for a decade. It is also improved many times according to requirements of sellers and customers. It does the same work that is calculating the bill, gives it to the customer and maintain proper database. They are accurate in calculation and printing, they also generate records. A new concept is also added in the billing system is thatthey also maintain relationships with the customers who purchase more products from the store regularly. System also concerns their requirements and gives them more commission. It also shows the overall profit and profit on aparticular product and give repots which items are required and which have cross their expiry date.

Bottlenecks of the Existing System :

Every system has pros and cons so existing system also have many advantages and disadvantages. So the bottlenecks of the existing system are as follows:

1. **User Interface:** User Interface is not so much good that operators feel some problems in working.
2. **Graphical User Interface:** GUI is not good so the operators get bored by watching screen.
3. **Processing speed:** Processing speed of the software is not so much good to operate fast.
4. **Flexible:** Existing system is not so much flexible that can be changed according to the operators and customers.
5. **Automatic generation of the Reports:** Not able to automatically generate the reports and documents.
6. **Workload:** Sometimes the system hangs when workload is more.
7. **Error Free:** Sometimes the system gives error in the calculation in making the bill and in the information of the products due to workload.
8. **Man Power:** Existing system uses so many people to operate the system.
9. **Resources:** System does not use the resources properly.

Proposed System

The proposed system is intended to provide the facility of automating the billingsystem for the supermarket. To reduce the bottlenecks of the existing system there is a need to develop a new system. The new system should concern the requirements of the customer and the sellers. This project is designed with a goal to making the existing system more informative, reliable, fast and easier. There are many reasons for the starting of the project because in the selling of items through the manual system of salesperson faces a lot of inefficiencies. Itrequires handling of large record books that consist of both irrelevant and important information’s thus making it difficult to find out the required information as per necessity. Object Oriented Programming concepts enabled me to implement the project modules in a easier way.

Advantages of the proposed system:

- Converts the manual work which is time consuming and error prone to fully automated system.
- Helps in eliminating the paper work ,saves time and improves customer services.
- Makes the addition of items in the bill, deletion of items and modification of items in the bill easier and faster.
- C++ has supported to implement object oriented concepts.
- Bills can be calculated more easily and with more accuracy.
- Reduces pressure on the labor.
- Makes the system more feasible and flexible and thus retrieval of information becomes convenient.

Object Oriented Programming:

The major motivating factor in the invention of object-oriented approach is to remove some of the flaws encountered in the procedural approach. OOP treats data as a critical element in the program development and does not allow it to flow freely around the system. It ties data more closely to the function that operate on it, and protects it from accidental modification from outside function.OOP allows decomposition of a problem into a number of entities called objectsand then builds data and function around these objects. The data of an object can be accessed only by the function associated with that object.However, function of one object can access the function of other objects.

Some of the features of object-oriented programming are:

- Emphasis is on data rather than procedure.
- Programs are divided into what are known as objects.
- Data structures are designed such that they characterize the objects.
- Functions that operate on the data of an object are ties together in the data structure.
- Data is hidden and cannot be accessed by external function.

• Objects may communicate with each other through function.

• New data and functions can be easily added whenever necessary.

• Follows bottom up approach in program design. Object-oriented programming is the most recent concept among programming paradigms and still means different things todifferent people.

Basic Concepts of Object Oriented Programming :

It is necessary to understand some of the concepts used extensively in object-oriented programming.

These include:

- Objects
- Classes
- Data abstraction and encapsulation
- Inheritance
- Polymorphism
- Dynamic binding
- Message passing

Objects:

Objects are the basic run time entities in an object-oriented system. They may represent a person, a place, a bank account, a table of data or any item that the program has to handle. They may also represent user-defined data such as vectors, time and lists. Programming problem is analyzed in term of objects and the nature of communication between them. Program objects should be chosen such that they match closely with the real-world objects. Objects take up space in the memory and have an associated address like a record in Pascal, or a structure in c.

Classes:

We just mentioned that objects contain data, and code to manipulate that data. The entire set of data and code of an object can be made a user-defined data type with the help of class. In fact, objects are variables of the type class. Once a class has been defined, we can create any number of objects belonging to that class. Each object is associated with the data of type class with which they are created. A class is thus a collection of objects similar types.

Data Abstraction and Encapsulation:

The wrapping up of data and function into a single unit (called class) is known as encapsulation. Data and encapsulation is the most striking feature of a class. The data is not accessible to the outside world, and only those functions which are wrapped in the class can access it. These functions provide the interface between the object's data and the program. This insulation of the data from direct access by the program is called data hiding or information hiding. Abstraction refers to the act of representing essential features without including the background details or explanation. Classes use the concept of abstraction and are defined as a list of abstract attributes such as size, wait, and cost, and function operate on these attributes. They encapsulate all the essential propertiesof the object that are to be created. The attributes are sometime called data members because they hold information. The functions that operate on thesedata are sometimes called methods or member function.

Inheritance:

In object-oriented programming, inheritance is the mechanism of basing an object or class upon another object or class , retaining similar implementation. In most class-based object-oriented languages, an object created through inheritance (a "child object") acquires all the properties and behaviors of the parent object (Except: constructors, destructor, overloaded

operators and friend functions of the base class). Inheritance allows programmers to create classes that are built upon existing classes,[1] to specify a new implementation while maintaining the same to reuse code and to independently extend original software via public classes and interfaces. The relationships of objects or classes through inheritance give rise to a directed graph.

An inherited class is called a subclass of its parent class or super class. The term "inheritance" is loosely used for both class-based and prototype-based programming, but in narrow use the term is reserved for class-based programming (one class inherits from another), with the corresponding technique in prototype-based programming being instead called delegation (one object delegates to another).

Inheritance should not be confused with sub typing. In some languages inheritance and sub typing agree, whereas in others they differ in general, sub typing establishes an is a relationship, whereas inheritance only reuses implementation and establishes a syntactic relationship, not necessarily a semantic relationship (inheritance does not ensure behavioural). To distinguish these concepts, sub typing is also known as interface inheritance, whereas inheritance as defined here is known as implementation inheritance or code inheritance. Still, inheritance is a commonly used mechanism for establishing subtype relationships.

Polymorphism:

Polymorphism is another important OOP concept. Polymorphism, a Greek term, means the ability to take more than on form. An operation may exhibit different behaviour is different instances. The behaviour depends upon the types of data used in the operation. For example, consider the operation of addition. For two numbers, the operation will generate a sum. If the operands are strings, then the operation would produce a third string by concatenation. The process of making an operator to exhibit different behaviours in different instances is known as operator overloading.

Dynamic Binding :

Binding refers to the linking of a procedure call to the code to be executed in response to the call. Dynamic binding means that the code associated with a given procedure call is not known until the time of the call at run time. It is associated with polymorphism and inheritance. A function call associated with a polymorphic reference depends on the dynamic type of that reference.

Message Passing :

An object-oriented program consists of a set of objects that communicate with each other. The process of programming in an object-oriented language, involves the following basic steps:

1. Creating classes that define object and their behavior
2. Creating objects from class definitions
3. Establishing communication among objecct

Objects communicate with one another by sending and receiving information much the same way as people pass messages to one another. The concept of message passing makes it easier to talk about building systems that directly model or simulate their real world counterparts.

File Concepts:

The I/O system of C++ handles file operations which are very much similar to the console input and output operations .It uses file streams as an interface between the programs and files. The stream that supplies data to the program is called input stream and the one that receives data from the program is called output stream. In other words input stream extracts data from the file and output stream inserts data to the file. The input operation involves the creation of an input stream and linking it with the program and input file. Similarly, the output operation involves establishing an output stream with the necessary links with the program and output file.

Detail of file stream classes:

- filebuf :Its purpose is to set the file buffers to read and write. Contains Open prompt constant used in the open() of file stream classes. Also contain close() and open() as members.
- fstreambase: Provides operations common to file streams. Serves as a base for fstream, ifstream and ofstream class. Contains open() and close() functions.
- ifstream: Provides input operations. Contains open() with default input mode. Inherits the functions get(),getline(),read(),seekg(),tellg() functions from istream.
- ofstream: Provides output operations. Contains open() with default output mode. Inherits put(),seekp(),tellp() and write() functions from ostream.
- fstream: Provides support for simultaneous input and output operations.

- Contains open with default input mode. Inherits all the functions from istream and ostream classes through iostream.

File Mode Operation:

Parameter	Meaning
ios::app	Append to end-of-file
ios::ate	Go to end-of-file on opening
ios::binary	Binary file
ios::in	Open file for reading only
ios::nocreate	Open fails if file the file does not exists
ios::noreplace	Open fails if the file already exists
ios::out	Open file for writing only
ios::trunk	Delete the contents of the file if it exists

CHAPTER-2
SYSTEM SPECIFICATIONS

2.2 HARDWARE SPECIFICATIONS

Processor	:	Intel Core i5 Processor
RAM	**:**	**4GB(Minimum)**
HDD	:	40GB(Minimum)

2.3 SOFTWARE SPECIFICATIONS

Operating System	:	Windows 11
Front-end	:	VSC
Languages Used	**:**	**C++**

CHAPTER 3

MODULE DESCRIPTION

There are 5 modules in this project. These are as follows:

1) Login -Id

2) Apply taxes and commissions

3)Check the report

4)Enter the information about products

5)Calculate the bill

Module 1: Login–Id: This module is made for the login of users. We know that we have 3 users so login-id is for:

1. Administrator

2. Data Entry Operator

3. Bill Calculating Operator

Module 2: Apply taxes and commissions

This module is for administrator who will

1. Set the taxes for the products.

2. Set the commissions for the products.

Module 3: Check the Report

This module is also for the administrator who can generate or check the report of the product and how many products are sold on particular date or in a period of time.

Module 4: Enter the information about products

This module is for data entry operator who will

1. Enter which products come in the store.

2. Prices and expiry date of the product.

Module-5: Calculate the bill

This module is for bill calculating operator who will

1. Calculate the bill.

2. Print it.

CHAPTER 4

CODING

```
#include <iostream>
#include <fstream>
using namespace std;
class shopping
{
    private:
        int pcode;
        float price;
        float dis;
        string pname;
    public:
        void menu();
        void administrator();
        void buyer();
        void add();
        void edit();
        void rem();
        void list();
        void receipt();

};
void shopping::menu()
{
    m:
    int choice;
    string email;
    string password;
    cout<<"\t\t\t\t_____________________________________________\n";
    cout<<"\t\t\t\t_____________________________________________\n";
    cout<<"\t\t\t\t                 Supermarket main menu       \n";
    cout<<"\t\t\t\t_____________________________________________\n";
    cout<<"\t\t\t\t_____________________________________________\n";
    cout<<"\t\t\t\t|  1) Administrator     |\n";
    cout<<"\t\t\t\t|                       |\n";
    cout<<"\t\t\t\t|  2) Buyer             |\n";
    cout<<"\t\t\t\t|                       |\n";
    cout<<"\t\t\t\t|  2) Exit              |\n";
    cout<<"\n\t\t\t     Please Select!";
    cin>>choice;

    switch (choice)
    {
        case 1:
        cout<<"\t\t\t Please login \n";
```

```
        cout<<"\t\t\t Enter email   \n";
        cin>>email;
        cout<<"\t\t\t Password      \n";
        cin>>password;

        if(email=="robby@email.com" && password=="robby@123")
        {
            administrator();
        }
        else{
            cout<<"Invalid email/password";

        }
        break;

    case 2:
    {
        buyer();
    }
    case 3:
    {
        exit(0);
    }
    default:{
        cout<<"Please select from the given options";
    }
    goto m;}
}


void shopping:: administrator()
{
    m:
    int choice;

    cout<<"\n\n\n\\t\t\t Administrator menu";
    cout<<"\n\t\t\t_______________1)Add the product_____";
    cout<<"\n\t\t\t                                     ";
    cout<<"\n\t\t\t_______________2)Modify the product_____";
    cout<<"\n\t\t\t                                     ";
    cout<<"\n\t\t\t_______________3)Delete the product_____";
    cout<<"\n\t\t\t                                     ";
    cout<<"\n\t\t\t_______________4) Back to main menu_____";

    cout<<"\n\t\t\t Please enter your choice";
    cin>>choice;

    switch(choice)
    {
        case 1:
            add();
            break;
        case 2:
            edit();
            break;
```

```
        case 3:
            rem();
            break;
        case 4:
            menu();
            break;
        default:
        cout<<"Invalid choice!";
    }
    goto m;

}
void shopping:: buyer()
{
    m:
    int choice;
    cout<<"\t\t\t  Buyer \n";
    cout<<"\t\t\t                \n";
    cout<<"\t\t\t1) Buy product\n";
    cout<<"\t\t\t                \n";
    cout<<"\t\t\t2) Go back      \n";
    cout<<"\t\t\t Enter your choice: ";
    cin>>choice;

    switch(choice)
    {
        case 1:
            receipt();
            break;
        case 2:
            menu();
        default:
            cout<<"Invalid choice";
    }
    goto m;

}

void shopping::add()
{
    m:
    fstream data;

    int c;
    int token=0;
    float p;
    float d;
    string n;
    cout<<"\n\n\t\t\t Add new product";
    cout<<"\n\n\t\t\t Product code of the product";
    cin>>pcode;
    cout<<"\n\n\t\t\t Name of the product";
    cin>>pname;
    cout<<"\n\n\t\t\t Price of the product";
```

```
    cin>>price;
    cout<<"\n\n\t\t\t Discount on product";
    cin>>dis;

    data.open("database.txt",ios::in);

    if(!data)
    {
        data.open("database.txt",ios::app|ios::out);
        data<<"  "<<pcode<<"  "<<pname<<"  "<<price<<"  "<<dis<<"\n";
        data.close();
    }
    else{
        data>>c>>n>>p>>d;

        while(!data.eof())
        {
            if(c==pcode)
            {
                token++;

            }
            data>>c>>n>>p>>d;
        }
        data.close();
    }
    if(token==1)
        goto m;
    else{
        data.open("database.txt",ios::app|ios::out);
        data<<"  "<<pcode<<"  "<<pname<<"  "<<price<<"  "<<dis<<"\n";
        data.close();

    }
    cout<<"\n\n\t\t\t Record Inserted";

}
void shopping::edit()
{
    fstream data,data1;
    int pkey;
    int token=0;
    int c;
    float p;
    float d;
    string n;

    cout<<"\n\n\n\t\t Modify the record";
    cout<<"\n\n\n\t\t Product code:";
    cin>>pkey;

    data.open("database.txt",ios::in);
    if(!data)
    {
```

```
        cout<<"\n\nFIle doesnt exist";
    }
    else{
        data1.open("database1.txt", ios::app|ios::out);

        data>>pcode>>pname>>price>>dis;
        while(!data.eof())
        {
            if(pkey==pcode)
            {
                cout<<"\n\n\t\t Product new code:";
                cin>>c;
                cout<<"\n\n\t\t Name of the product";
                cin>>n;
                cout<<"\n\n\t\t price : ";
                cin>>p;
                cout<<"\n\n\t\t Discount: ";
                cin>>d;
                data1<<"  "<<c<<" "<<n<<" "<<p<<" "<<d<<"\n";
                cout<<"\n\n\n\t\t\t Record edited ";
                token++;

            }
            else {
                data1<<" "<<pcode<<" "<<pname<<" "<<price<<" "<<dis<<"\n";
            }
            data>>pcode>>pname>>price>>dis;
        }
        data.close();
        data1.close();

        remove("database.txt");
        rename("database1.txt","database.txt");

        if(token==0)
        {
            cout<<"\n\n\t\t Record not found sorry!";
        }
    }
}
void shopping::rem()
{
    fstream data,data1;
    int pkey;
    int token=0;
    cout<<"\n\n\n\t Delete Product";
    cout<<"\n\n\n\t Product code";
    cin>>pkey;
    data.open("database.txt",ios::in);
    if(!data)
    {
        cout<<"File doesnt exist";
    }
    else{
```

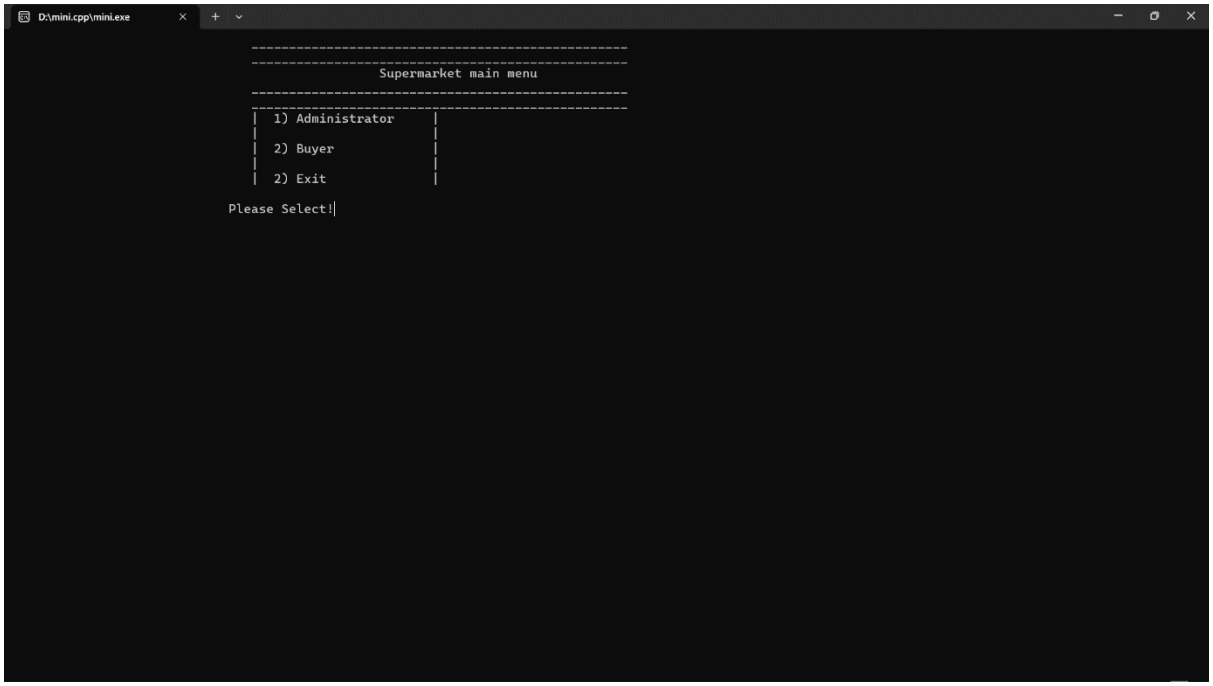
```
        data1.open("database1.txt",ios::app|ios::out);
        data>>pcode>>pname>>price>>dis;
        while(!data.eof())
        {
            if(pcode==pkey)
            {
                cout<<"\n\n\t Product deleted succesfully";
                token++;
            }
            else{
                data1<<" "<<pcode<<" "<<pname<<" "<<price<<" "<<dis<<"\n";
            }
            data>>pcode>>pname>>price>>dis;
        }
        data.close();
        data1.close();
        remove("database.txt");
        rename("database1.txt","database.txt");

        if(token==0)
        {
            cout<<"\n\n\n Record not found";
        }
    }
}

void shopping::list()
{
    fstream data;
    data.open("database.txt",ios::in);
    cout<<"\n\n___________________\n";
    cout<<"ProNo\t\tName\t\tPrice\n";
    cout<<"\n\n___________________\n";
    data>>pcode>>pname>>price>>dis;
    while(!data.eof())
    {
        cout<<pcode<<"\t\t"<<pname<<"\t\t"<<price<<"\n";
        data>>pcode>>pname>>price>>dis;
    }
    data.close();
}
void shopping::receipt()
{
    m:
    fstream data;
    int arrc[100];
    int arrq[100];
    string choice;
    int c=0;
    float amount=0;
    float dis=0;
    float total=0;

    cout<<"\n\n\t\t\t RECEIPT";
```

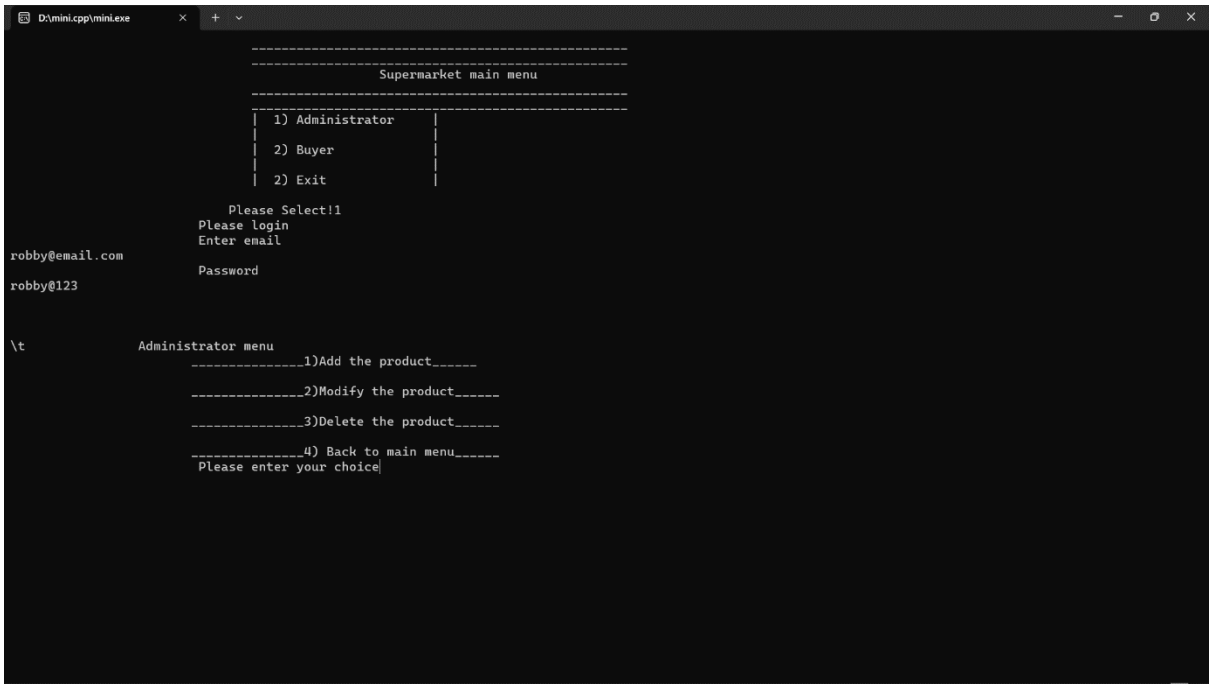
```
    data.open("database.txt",ios::in);
    if(!data)
    {
        cout<<"n\n\n Empty database";
    }
    else{
        data.close();

        list();
        cout<<"\n_____________________________________\n";
        cout<<"\n_____________________________________\n";
        cout<<"\n            Place the order            \n";
        cout<<"\n_____________________________________\n";
        cout<<"\n_____________________________________\n";
        do
        {
            cout<<"\n\n Enter product code:";
            cin>>arrc[c];
            cout<<"\n\nEnter the product quantity:";
            cin>>arrq[c];
            for(int i=0;i<c;i++)
            {
                if(arrc[c]==arrc[i])
                {
                    cout<<"Duplicate product code. Please try again!";
                    goto m;
                    }
                }
                c++;
                cout<<"\n\n Do you want to buy another product? If yes then
press y else no";
                cin>>choice;
            }
            while(choice =="y");
            cout<<"\n\n\n\t\t_____________________RECEIPT_________________\n";
            cout<<"\nProduct no\t Product name \t product quantity\t price\t
Amount\t Amount with discount\n";

            for(int i=0;i<c;i++)
            {
                data.open("database.txt",ios::in);
                data>>pcode>>pname>>price>>dis;
                while(!data.eof())
                {
                    if(pcode==arrc[i])
                    {
                        amount=price*arrq[i];
                        dis=amount-(amount*dis/100);
                        total=total+dis;
                        cout<<"\n"<<pcode<<"\t\t"<<pname<<"\t\t"<<arrq[i]<<"\t\t
"<<price<<"\t\t"<<amount<<"\t\t\t"<<dis;
                    }
                    data>>pcode>>pname>>price>>dis;
                }
```

```
            }
            data.close();
        }
        cout<<"\n\n_____________________________________";
     cout<<"n\n Total amount : "<<total;
 }


int main()
{
    shopping s;
    s.menu();
}
```

CHAPTER 5

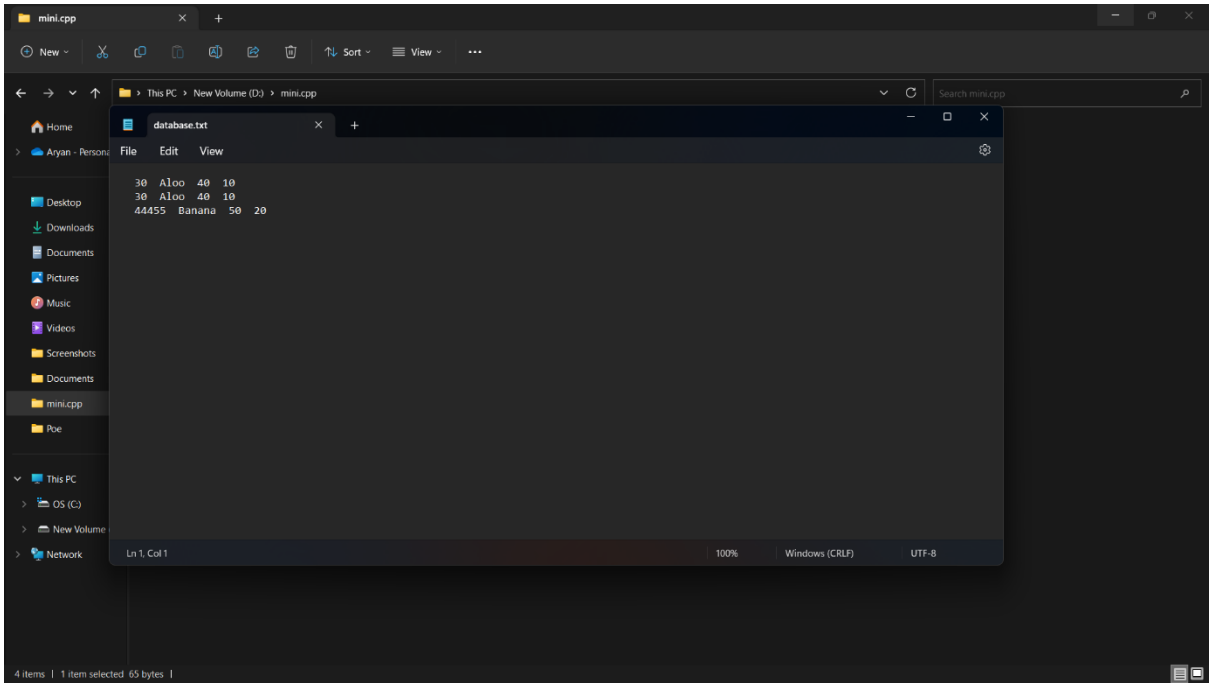
SCREEN SHOTS





```
D:\mini.cpp\mini.exe

\t              Administrator menu
                      ______________1)Add the product_____

                      ______________2)Modify the product_____

                      ______________3)Delete the product_____

                      ______________4) Back to main menu_____
                       Please enter your choice1


                       Add new product

                       Product code of the product44455


                       Name of the productBanana


                       Price of the product50


                       Discount on product

20


                       Record Inserted


\t              Administrator menu
                      ______________1)Add the product_____

                      ______________2)Modify the product_____

                      ______________3)Delete the product_____

                      ______________4) Back to main menu_____
                       Please enter your choice
```

```
D:\mini.cpp\mini.exe
ProNo           Name            Price


-------------------
30              Aloo            40
30              Aloo            40
44455           Banana          50

---------------------------------------

---------------------------------------

            Place the order

---------------------------------------

---------------------------------------

 Enter product code:44455


Enter the product quantity:4


 Do you want to buy another product? If yes then press y else nono


                  _______________________RECEIPT_______________

Product no       Product name    product quantity        price   Amount  Amount with discount

44455           Banana          4               50              200                     160

_____________________________________n
 Total amount : 160                     Buyer

                       1) Buy product

                       2) Go back
                        Enter your choice:
```


mini.cpp
New
Sort
View
This PC › New Volume (D:) › mini.cpp
Search mini.cpp
Home
Aryan - Persona
Desktop
Downloads
Documents
Pictures
Music
Videos
Screenshots
Documents
mini.cpp
Poe
This PC
OS (C:)
New Volume
Network
database.txt
File
Edit
View
30 Aloo 40 10
30 Aloo 40 10
44455 Banana 50 20
Ln 1, Col 1
100%
Windows (CRLF)
UTF-8
4 items | 1 item selected 65 bytes |

CHAPTER 6

CONCLUSION

This software has been developed designed to reduce the time taken to handle the sales activity. It is designed to replace an existing manual record system for reducing time taken for calculations and for storing data. This system has been developed with oops concepts.The system is strong to handle daily operations where the database is cleared over certain time. This system will reduce manual work, calculations and will also provide periodic reports any time.